The Ties that Bind: Growing Up with Immigrants and Later Life Attitudes toward Immigration

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Abstract:

In recent years, inequalities in residential environments have received much attention as an explanation for the appeal of anti-immigration parties across advanced democracies. This paper explores the long-term association between neighborhood and school context in adolescence and attitudes towards immigration later in life, drawing on a UK-based cohort study matched with fine-grained census data on immigrant composition across both sites of contact. Drawing on the premises of the impressionable years hypothesis, neighbourhood effects, and contact theory, we posit that as a space of social interaction and socialisation, the presence of immigrants in the local area that one experiences in adolescence is influential to the development of positive immigration attitudes in adulthood. Our findings suggest that the share of immigrants in one's neighbourhood and school during adolescence has a persistent positive association with favorable attitudes toward immigration among White British respondents 18 years later, net of parental characteristics in adolescence and individual and residential characteristics in adulthood. Further, this positive effect is robust in an instrumental variable design using an immigrant enclave shift-share instrument. These findings highlight the role of early-years socialization experiences in the formation of out-group attitudes and the importance of inter-group contact for fostering tolerant societies.

Keywords: immigration attitudes, political attitudes, neighborhood effects, contact theory, impressionable years

Introduction

Residential environments have long been highlighted as a major explanatory factor for understanding attitudes toward immigration across advanced democracies. A substantial body of research illustrates that where one lives shapes immigration attitudes above and beyond individual-level factors, such as educational attainment or employment status (Arzheimer et al., 2024; Bolet, 2020; 2021; Colantone & Stanig, 2018; Cools et al., 2021; Enos, 2017; Hangartner et al., 2019; Vasilopoulos et al., 2022). Existing literature emphasizes three mechanisms linking residential environments to attitudes. The first concerns local economic conditions. Research has shown that different forms of local deprivation, broadly conceived, are associated with anti-immigrant attitudes and support for far right parties (Arzheimer et al., 2024; Colantone & Stanig, 2018; Cremaschi et al., 2024; Vasilopoulos et al., 2022). Second, the ethnoracial composition of residential spaces, e.g. local shares of immigrants and minorities, has been shown to influence attitudes toward immigration among natives (Claassen & McLaren, 2022; Enos, 2017; Hangartner et al., 2019; Patana, 2020; Vasilopoulos et al., 2022). Finally, a third group of studies illustrates that social deprivation, involving status decline, depopulation, and loss of sociocultural spaces, is associated with decreased tolerance toward immigrants (Bolet, 2021; Sanchez-Garcia et al., 2025; Stroppe, 2023).

A common feature of this existing research is that it focuses on the influence of voters' current residential environments on immigration attitudes in adulthood. This approach is theoretically justified. For instance, studies often draw on the premises of Realistic Group Conflict Theory (Sherif et al., 1961) to argue that interethnic animus increases when natives and immigrants compete for jobs or access to welfare services (Bolet, 2020; Halikiopoulou & Vlandas, 2020). Similarly, the ethnic threat hypothesis posits that natives react with hostility to immigrants shortly following an increase in the share of foreigners in an area (Enos, 2017;

Hangartner et al., 2019).

On the other hand, very little is known about whether residential characteristics experienced earlier in the life cycle affect political attitudes later in life. Is the influence of the residential environment on immigration attitudes entirely limited to one's extant local circumstances? Or does the local context experienced during one's formative years exert an enduring influence above and beyond current conditions? This article aims to address this gap. Drawing on the premises of the impressionable years hypothesis, neighborhood effects and contact theory, we investigate whether growing up in local areas with increased immigrant presence carries a long-term influence on attitudes toward immigration later in life. To explore this question, we draw on a cohort study from the United Kingdom, *Next Steps*. The panel follows a large sample of adolescents into adulthood over a period of 18 years. Local immigrant composition in adolescence is measured by merging Next Steps with census data at the neighborhood level (LSOA), as well as at the level of the school attended at age 14. Our findings suggest that White British respondents who lived in neighborhoods with higher shares of immigrants during adolescence hold more favorable views of immigration nearly two decades later. This relationship persists after controlling for parental characteristics in adolescence as well as socioeconomic and residential characteristics in adulthood. We also observe this long-term positive relationship when school-level immigrant composition is used. Finally, estimates of the local immigrant share effect are also robust in instrumental variable models using an immigrant enclave shift-share instrument, suggesting that this relationship is not an artefact of residential selection decisions.

These findings make a number of important contributions. First, they advance a rich literature on the association between early life conditions and political behavior in adulthood - which has largely focused on national-level context (e.g. Laaker, 2024; McLaren et al., 2021) - by highlighting the influence of local conditions, both in neighborhoods and at

school, and in so doing, offer more robust empirical support for contact theory. Second, the article adds new evidence to an old debate in political socialization: are political attitudes formed during adolescence and show limited change in adulthood, as *life persistence* models suggest (e.g. Devine &Valgarðsson, 2024; Jennings & Niemi, 1978; Kustov et al., 2021)? Or do immigration attitudes continuously evolve by contemporaneous circumstances and experiences regardless of one's early life conditions as *life-long openness* models of political attitudes would anticipate? (Lersch, 2023). Third, the findings are important for understanding the long-term implications of increased ethnic diversity in Western advanced democracies for political competition and interethnic conflict. Given that growth in migration and ethnoracial populations at the national level translates into more diverse neighborhoods (Catney et al., 2023; Zhang and Logan 2025), more children and adolescents are brought up in increasingly immigrant-dense communities. Our findings suggest that this rise is likely to translate into long-term support - rather than hostility - toward immigrants among White populations.

Theoretical Framework

The Impressionable Years Hypothesis and Enduring Neighborhood Effects

Political science research has long investigated the existence of a critical period during adolescence and early adulthood when individuals' major political orientations are particularly malleable, only to change little thereafter (Dinas, 2010; Jennings & Niemi, 1978; Mannheim, 1952; Sears & Brown, 2023). Known as the impressionable years hypothesis, this framework argues that younger people have limited experience with the social and political world and, as a result, are more open to influence from pivotal political events and processes as well as from the formative aspects of their social environment (Jennings & Markus, 1984; Sears & Valentino, 1997). With ageing, political experience and knowledge tends to increase

while the influence of exogenous sociopolitical stimuli decreases, as new information is weighted against a richer repository of past information and experience (Jennings & Markus, 1984, p. 1001). In his classic work on political generations, Mannheim referred to this phenomenon as "stratification of experience," arguing that earlier impressions constitute what we perceive as the "natural view of the world", whereas later experiences "tend to receive their meaning from this original set, whether they appear as that set's verification and fulfillment or as its negation and antithesis" (Mannheim, 1952, p. 177).

The impressionable years hypothesis has received broad empirical support. A number of research projects show that, indeed, political attitudes fluctuate more widely in adolescence and young adulthood, compared to later in life (Alwin & Krosnick, 1991; Jennings & Niemi, 1978; Visser & Krosnick, 1998; Laaker, 2024). Longitudinal studies offer further support by showing that political orientations tend to exhibit limited change during mature adulthood (e.g. Alwin & Krosnick, 1991; Miller & Sears, 1986; Sears & Funk, 1999; Kustov et al., 2021; Peterson et al., 2020). Evidence of the impressionable years hypothesis has specifically been shown for attitudes toward immigration. In a comparative study drawing on panel data across democracies, Kustov and coauthors found that, among older adults, immigration attitudes remained fairly stable over time, while younger people were more likely to change preferences (2021).

At the same time, a large body of empirical studies across several national contexts points to long-term effects of the neighborhood experienced during childhood or adolescence on a range of individual outcomes measured later in life. While the neighborhood effects literature has predominately focused on the effects of neighborhood disadvantage on residential, socioeconomic, or health outcomes in adulthood (Sharkey 2008; Chetty et al. 2016; Ludwig et al. 2013), it highlights social-interactive processes (Galster 2011) that are relevant for explaining the development of political attitudes as well. Neighborhoods

represent key sites of socialization, governing exposure to peer groups, role models, social networks and predominant local norms, values and culture (Galster 2011; Sharkey and Faber 2014). As Galster (2011) argues, social contagion and collective socialization mechanisms within the local area mean that contact with peers in neighborhoods and the presence of role models conveying social norms are likely to influence attitude formation. As adolescents are particularly open to peer influence, these mechanisms are likely to be particularly salient at this period in their lives, forming the context in which political orientations take root and grow. Some extant empirical evidence suggests, for instance, that neighborhood environments experienced early in the life course have long-term effects on political outcomes such as electoral registration or political partisanship (McAvay and Vasilopoulos 2024; Brown et al. 2021).

Growing Up Near Immigrants: Contact vs Threat

How might neighbourhood immigrant composition during adolescence shape immigration attitudes? A vast number of studies have investigated how spatial proximity and interaction with minority outgroups shape immigration attitudes among the ethnic majority. Contact theory represents one of the oldest and most studied frameworks for understanding this relationship. Allport (1954) argued that under certain conditions, ethnic majority contact with minority outgroups can reduce hostility and prejudice. These conditions involve equal status between the group members within the context situation; the pursuit of common goals; intergroup co-operation without competition across group lines; and the contact being encouraged by law, custom, or authorities. Subsequent research has relaxed these assumptions showing that prejudice reduction can take place when one or more of the conditions is violated (Pettigrew & Tropp, 2006; Paluck et al., 2018). Contact is thought to reduce intergroup hostility by providing information about outgroup members that counters stereotypical depictions of ethnic minority groups and by cultivating empathy through

perspective-taking (Pettigrew & Tropp, 2008).

A number of empirical studies have provided support for contact theory. Research has repeatedly shown that sustained, durable, and equal status contact with ethnic minorities and outgroups translates into a reduction of prejudice (Binder et al., 2009; van Heerden & Ruedin, 2019; Khali & Naumann, 2022; Vezzali & Stathi, 2021; Vasilopoulos et al. 2022). Nonetheless, the large majority of these studies explore the contemporaneous relationship between local immigrant shares and attitudes in adulthood, ignoring potential long-term effects starting in the impressionable years. Two studies are exceptional in this regard. Focusing on the UK, McLaren and her colleagues found that younger generations who spent their formative years during periods of high (versus low) ethnic diversity at the national level were more supportive toward immigration compared to older cohorts. They attribute this finding to the "normalisation" of visible first and second generation immigrants through increasing presence in sports, culture, and the media as conducive to prejudice reduction (McLaren et al., 2021). In their analysis on the role of immigrant composition on attitudes later in life, Eger et al., draw on the foreign born populations in U.S. states during one's formative years, showing a positive effect later in life (2022), while a recent study investigating the same hypotheses using counties in Germany finds a small negative effect (Mazza et al., 2026).

Nonetheless, these studies do not measure effects of minority group presence at smaller spatial scales, which allows for a more precise measure of contact. Evidence of the positive effects of contact at a more granular level come instead from Brown et al. (2021), who show that, in the US, having an African American as an immediate next door neighbor during childhood had a lasting effect on Whites' political partisanship, rendering them more likely to identify with the Democratic party up to seven decades later.

However, no studies to our knowledge have explored the long-term influence of neighborhood composition during adolescence on attitudes towards immigration later in life. Attending the same school or participating in sports, play, or other neighborhood activities may well foster the type of contact that leads to a reduction in prejudice, through the social-interactive mechanisms highlighted by Galster (2011). Existing work shows that school-level factors such as intergroup friendships and a perceived multicultural and democratic educational environment matter to youth's immigrant attitudes, albeit without directly measuring school-level shares of immigrants (Eckstein et al. 2021; Miklikowska 2017; Wölfer et al. 2018). Further, close proximity with immigrants in neighborhoods and schools offers opportunities for relationship-building through friendship or couple formation (Rahnu et al. 2018; Puur et al. 2022). Finally, contact with immigrants provides direct insights into their lived experiences, beliefs, and contributions to the host society that may help counter stigmatizing stereotypes (Pettigrew & Tropp 2008).

Not all contact, however, fosters tolerance. Allport argues that the arrival of ethnic minorities in a vicinity may in fact activate ethnic threat (Allport, 1954), particularly when rapid influxes of immigrants occur in the broader vicinity, not allowing for sustained inter-group contact. Prejudice-reducing contact is more likely to take place in small spatial units, i.e. at the neighborhood level. Subsequent research demonstrates the conditions under which threat tends to be triggered. Sudden changes in foreigner numbers in the broader vicinity have repeatedly been shown to evoke negative attitudes toward immigration (Enos, 2017; Hangartner et al., 2019), while anti-immigrant sentiments run high in communities that are within traveling distance or part of broader regions with immigrant-dense areas (Dazey & Gay, 2025; Evans & Ivaldi, 2021; Vasilopoulos et al. 2022). Spatial arrangements in which outgroups are distant are deleterious for interethnic relationships because they raise awareness over the existence of the outgroup while at the same time limiting opportunities for

the types of contact that lead to prejudice reduction (Dazey & Gay, 2025; Enos, 2017; Giles & Ivaldi, 2021; Hargartner et al., 2019).

We do not dismiss the possibility that exposure to local immigration in adolescence could result in negative attitudes later in life, in line with ethnic threat frameworks. However, the sustained, close proximity fostered by the adolescent neighbourhood is arguably more likely to establish the conditions under which tolerance - rather than threat - would develop. Therefore, we anticipate that exposure to immigration at the neighborhood level during adolescence will have a lasting influence in attitudes toward immigration later in life. Following contact theory, we expect that more positive attitudes will be found among White British youth living in local areas with increased shares of immigrants (H1). We further expect this relationship to be observed at the level of the school, namely that shares of immigrants in the school will positively relate to more favorable attitudes toward immigration later in life (H2).

The Role of Residential Selection

Whether contact or ethnic threat are at play in the formation of long-term immigration attitudes, both perspectives, as well as the neighborhood effects literature, posit a direct, causal effect of the local environment. However, residential selection could nonetheless intervene in the observed relationship between local immigrant shares in adolescence and immigration attitudes later in life.

The first potential selection mechanism relates to parental self-selection into immigrant-dense areas. If parents who are more tolerant toward immigration, or have socioeconomic characteristics that correlate with tolerance, such as university degrees or high-skill occupations (Dražanová et al. 2024), are more likely to move into such neighborhoods in the first place, the association observed between their children's attitudes

and the residential environment may be due to family socialization effects rather than neighborhood effects per se (Miklikowska 2017). Similarly, less tolerant parents may move out when immigrant shares in the local area reach a tipping point, consistent with white flight dynamics underpinning residential segregation in the UK and other contexts (Card et al. 2008; Clark and Coulter 2015).

Second, the intergenerational reproduction of context (Sharkey 2008; Hedman and Van Ham 2021.; McAvay 2018), namely the tendency for adults to remain in the same neighborhoods or move into ones similar to those experienced in childhood, could also confound the relationship of interest. If children who grew up in immigrant-dense areas are more likely to end up in immigrant-dense neighborhoods as adults, and if those current contexts impact upon immigrant attitudes, neighborhood effects in adolescence may simply be a reflection of current neighborhood conditions. Finally, immigrants do not randomly select into residential locations. Mobility choices are constrained by structural factors related to housing and job markets, discrimination, as well as preferences for local ethnoracial composition (i.e. ethnic enclaves) (Lymperopoulou 2013; Mahony & Rowe 2026). All of these factors could potentially result in immigrants sorting into residential environments with more tolerant White populations, thus biasing our estimation of the immigrant share effect.

Therefore, our empirical strategy seeks to disentangle the role of residential selection from direct effects of the adolescent neighborhood. First, we account for parental socioeconomic characteristics during the respondent's adolescence and use higher-order geographic fixed effects. This means that we (a) control for factors that potentially determine residential selection among parents and (b) account for parental selection decisions at the level of these higher geographies. Second, we introduce both socioeconomic and neighborhood-level controls measured in adulthood to address current residential selection and the intergenerational transmission of context. Third, we complement our

neighbourhood-level analysis with school-level data. Burgess et al. (2015) demonstrate that academic quality and geographic proximity are the most important determinants of school choice in Britain, suggesting parental selection based on levels of immigration will be less of a concern in these models. Furthermore, the structured nature of the classroom environment makes it difficult for children to select out of contact with out-groups in school settings (Wölfer et al. 2018). Finally, we adopt an instrumental variable design using an immigrant enclave shift-share instrument to address the possibility of residential self-selection among immigrants.

Data and Methods

We use data from Next Steps, a high-quality, nationally representative English cohort study beginning in 2004. The initial sample consisted of 15,770 13-14 year-old children (in Year 9 of English secondary education) who were attending either state schools, private/independent schools, or pupil referral units at the time of this first wave. Respondents were then surveyed annually from 2004 until 2010 (wave 7) before again being surveyed in 2015 (wave 8, at age 24-25) and 2022 (wave 9, at age 31-32). 7,279 cohort members completed the 2022/wave 9 survey, providing an overall retention rate of 46.16% (for more information on the survey, see Adali et al. 2025; Fang-Wei Wu et al. 2024).[1] Our analysis uses data from waves 1 and 9, allowing us to assess the relationship between neighbourhood diversity during the adolescent years and immigration attitudes in adulthood, 18 years later. As our aim is to probe these attitudes among the ethnic majority group, we restrict the analysis to White British respondents (67% of the total wave 1 sample).

Our outcome of interest, immigration attitudes, is measured via a question which asks

[1] We conduct attrition analysis at wave 1 to understand how those who remain in the sample differ from those who drop out before wave 9. Whilst the two groups differ on some sociodemographic characteristics, we control for these characteristics to minimise concerns with bias (Arel-Bundock and Pelc, 2018). See supplementary materials section 1 for a discussion of this analysis.

respondents “would you say it is generally bad or good for Britain’s economy that migrants come to Britain from other countries?”. This is an 11-point scale from ‘Extremely Bad’ to ‘Extremely Good’, with higher scores equating to more positive attitudes toward immigration. This question is measured at wave 9 of the survey, so captures immigration attitudes in adulthood (specifically at age 31-32). We assume that this measure taps a general disposition toward or tolerance of immigrants rather than simply capturing assessments of the economic utility of the migrants. In supplementary materials section 2, we validate this assumption using alternative British data from the same time period, showing that economic attitudes to immigration strongly correlate with cultural attitudes to immigration and support for immigration in general and that all three load onto a single latent dimension of immigration attitudes.

The key explanatory variable is the percentage of the population born outside of the UK in each respondent’s neighbourhood during adolescence, which we term the neighbourhood immigrant share. We derive this measure using data from the 2001 UK Census, a comprehensive survey of the entire UK population which provides us with high-quality population data at fine-grained levels of geographic aggregation. We use data at the LSOA level, an administrative unit comprising between 400-1,200 households (1,000-3,000 individuals) and the smallest unit of analysis available in the data. There were 32,482 LSOAs in England in the 2001 Census. We then match this data with LSOA identifiers for each respondent at wave 1, to derive a measure of the neighbourhood immigrant share at age 13-14 for each sample respondent. As our design relies on the assumption that geographic proximity is a good proxy for contact, this fine-grained measure is essential to capture this proximity and constitutes an improvement upon prior studies that have used much larger geographic units to proxy contact (Eger et al. 2022; Mazza et al. 2026). We address possible concerns that respondents may have relocated between 2001

(when Census data used to derive our immigrant share measure was collected) and 2004 (when Next Steps wave 1 was fielded) by restricting the model sample to respondents who have studied at the same school since 2001 (based on the assumption that an absence of school change is likely to imply stability in residential context). This drops around 500 respondents from our models, leaving us with a sample between 2800 and 2900 respondents, depending on the model.

We begin by assessing the relationship between neighborhood exposure to immigration in adolescence and immigration attitudes in adulthood. To do so, we estimate OLS regressions of immigration attitudes, measured in adulthood, on the adolescent neighbourhood immigrant share. Models include local authority fixed effects, meaning our estimates are derived from comparisons between individuals living in the same local authority, which had an average (median) population of 116,452 in 2004. These models therefore rule out residential sorting and time-invariant unobserved confounding at the local authority level. We control for gender, parental education, parental homeownership, parental household income and parental social class to try and account for parental socialization and the possibility of within-local authority selection effects among the parents of our respondents. To account for alternative possible effects related to neighborhood socioeconomic status, we also control for the LSOA-level unemployment rate and LSOA-level population density (using data from the 2001 Census), as immigrants often cluster in more urban and more economically disadvantaged neighbourhoods in Britain (Catney 2016; Zuccotti 2019).

In a second model, we add individual- and neighbourhood-level controls in adulthood (i.e. measured at wave 9): education, household income and the respondent's neighbourhood-level immigrant share in adulthood (using an LSOA-level measure derived

from the 2021 Census, measured one year before Next Steps wave 9). Whilst these latter controls are potentially post-treatment to adolescent exposure to immigrants, we believe they are valuable in allowing us to separate out the long-term influence of adolescent exposure from contemporaneous effects in adulthood. For a detailed breakdown of how all covariates are measured, see supplementary materials section 3. All models include weights to adjust for the complex sampling design in Next Steps (Fang-Wei Wu et al. 2024) and nonresponse bias (for a discussion of attrition and the calculation of non-response weights in Next Steps, see Silverwood et al. 2020). Standard errors are clustered by local authority and primary sampling unit.

This neighbourhood-level analysis has two potential limitations. First, whilst the residential environment is likely a good proxy for contact, we cannot be sure that the local neighbourhood was the primary site of contact and socialization for survey respondents. Second, whilst LSOAs are treated as independent from each other in the above models, unobserved spillover effects are possible if individuals socialize not just with residents of their own neighbourhood, but those of other neighbourhoods too. To address these concerns, we conduct further tests using an alternative site of contact - the school environment.

The school setting has been recognised as an ideal context in which to model the effects of intergroup contact in childhood and adolescence, because schools facilitate meaningful, frequent and sustained interactions with out-group members (Bohman and Miklikowska 2021). Additionally, as discrete environments with fixed boundaries, a school-level measure limits concern with spillover effects. Finally, given the modifiable areal unit problem and the recognition that observed relationships can vary depending on the level of geography used to measure constructs (Lee et al. 2025), a school-level test provides a means of validating that the relationship between adolescent contact and adulthood attitudes

holds across contexts and measures.

Given these advantages, we use the school-level data in Next Steps to assess the relationship between exposure to immigration at school and our outcome variable. Our key explanatory variable in these models is the percentage of the school population for whom English is not their first language (we do not have school-level data on country of birth, but this provides a solid proxy). Models again include local authority fixed effects and control for gender, parental education, parental homeownership, parental household income and parental social class. We add school-level controls - the percentage of pupils eligible for free school meals and the KS3 value added score (a measure of school performance) - to separate out intergroup contact from the long-term socioeconomic impacts of school performance or peer effects (e.g. Chetty et al. 2014; Sacerdote 2011) which could influence immigration attitudes in adulthood.

In a second model, we add individual- and neighbourhood-level controls in adulthood (i.e. measured at wave 9): education, household income and the respondent's neighbourhood immigrant share (again measured at the LSOA level in the 2021 Census). All models include non-response weights and standard errors are clustered by local authority and school.

An additional threat to identification is selection decisions among immigrants - if immigrants select into residential areas on the basis of local conditions that also influence the immigration attitudes of local (White British) residents (either our cohort members or their parents), coefficient estimates will be upwardly biased. To address these concerns, we assess the effect of neighbourhood context on attitudes to immigration using a variation of the immigrant enclave shift-share instrument developed by Card (2001). Using historic local immigrant shares at the local authority district level from the 1961 Census, we construct an instrument which predicts immigrant inflows from 1961 to 2001 in each district (i) for

migrant origin group *o* as the product of 1961 district-level shares and national-level shifts in the population of *o* from 1961 to 2001. We then sum across all origin groups *o* to give total predicted immigrant in-flows in each district *i* in 2001:

$$\Delta M_{i,2001} = \sum_{o} \left(\frac{M^{o}_{i,1961}}{M^{o}_{1961}} \right) \Delta M^{o}_{2001}$$

We then divide these counts by (predicted) district-level total population to convert them into district-level shares, and use these predicted shares as an instrument for actual immigrant shares in the local authority districts our cohort respondents reside in during adolescence in a 2SLS regression. For a detailed discussion of instrument construction and overview of instrument strength, see appendix section 6.1. The relevance of these historic shares as an instrument rests on the assumption that the residential choices of immigrants – and thus the neighbourhood-level immigrant shares in 2001 – are partially determined by historical settlement patterns (with new immigrants attracted to areas where their compatriots already live). In other words, as immigrants in Britain often choose to settle in neighbourhoods where their compatriots already reside (Lymperopoulou 2013), these historic local shares can be thought of as weights that determine exposure to the common shock (i.e. the national-level immigrant in-flows) (Goldsmith-Pinkham et al. 2020).

By using this instrument we seek to isolate the variance in 2001 immigrant shares that is determined by historic settlement patterns from that determined by contemporaneous local factors. These contemporaneous factors – such as local labour market conditions, housing availability or the (perceived) tolerance of local residents toward immigrants – could both attract immigrants and influence the attitudes of our cohort members (either directly or via parental socialization), biasing OLS estimates. In contrast, the 40-year lag between our historic shares and 2001 measures makes it less likely that our instrument will be biased by these contemporaneous factors.

To increase the plausibility of the identification assumption – that the historic shares are uncorrelated with unobserved determinants of immigration attitudes (Goldsmith-Pinkham et al. 2020) – we adopt a number of strategies. First, we again control for parental characteristics in adolescence (parental education, parental homeownership, parental household income and parental social class) to account for concerns with parental selection effects, and add respondent education and income, measured in adulthood, to account for persistent SES effects. Second, to account for potential endogeneity caused by persistent local characteristics (e.g. long-term political cultures or contemporary economic outcomes determined by historic occupational structures), all models include region fixed effects (a higher-order geographic unit), meaning that we account for persistent confounding at the regional level. Third, we estimate additional models that control for district-level correlates of the historic immigrant shares, also measured in the 1961 Census. We again apply survey weights to all models and cluster standard errors at the district level.

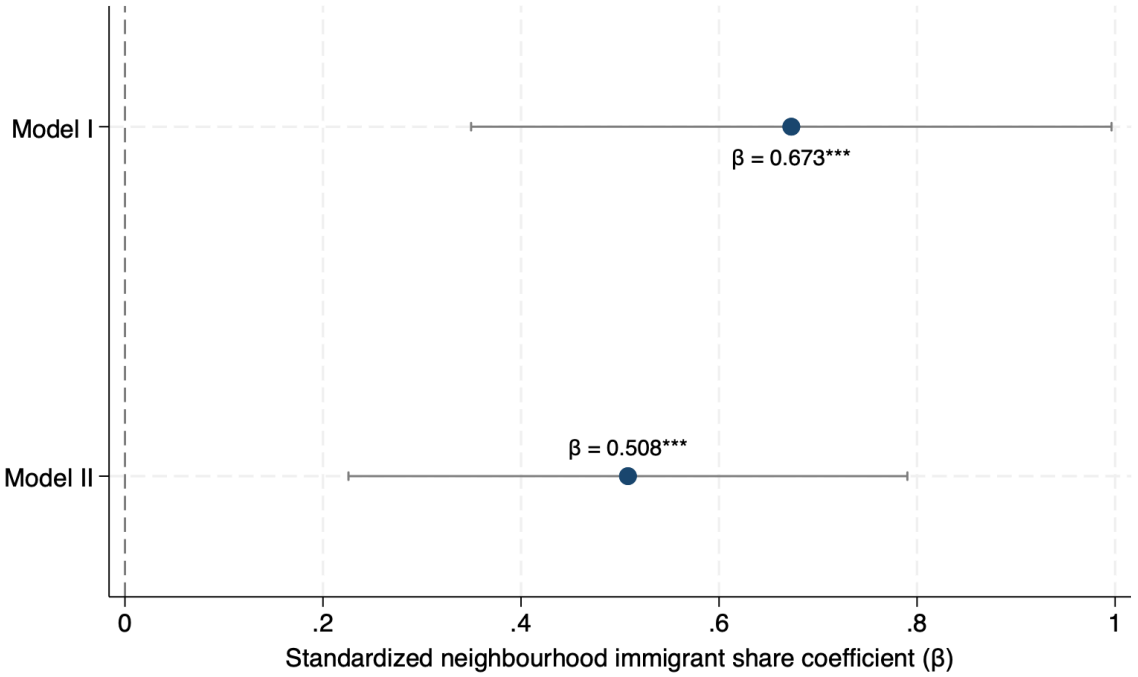
Results

Neighbourhood Analysis

Figure 1 shows the association between neighborhood exposure to immigration in adolescence and immigration attitudes in adulthood. Complete results are presented in supplementary materials table 4.1. The adolescent immigrant share variable is beta-standardized prior to inclusion in the model, meaning that the coefficient represents the change in support for immigration associated with an increase of 5.94 pp in the LSOA immigrant share. Across both models, we find support for H1 - the neighbourhood-level immigrant share is positively associated with support for immigration (the belief that immigration is good for the economy). The magnitude of this association is meaningful. Based on predicted outcome values derived from Model II, an individual who lives in an

age-14 neighbourhood at the 90th percentile for immigrant share (11.56%) is predicted to be 0.39 points (on the 0-10 outcome scale) more supportive of immigration in adulthood than an individual who lives in an age-14 neighbourhood at the 10th percentile for immigrant share (1.85%). An individual who resides in an extremely immigrant-dense neighbourhood at age 14 (the 99th percentile, or an immigrant share of 32.43%) is predicted to be 1.25 points more supportive of immigration in adulthood than one in a 10th percentile neighbourhood at age 14.

Figure 1: The regression of support for immigration on adolescent neighbourhood immigrant share



Notes: The plot reports coefficient estimates and 95% CIs. Model I controls for gender, neighbourhood characteristics and parental controls in adolescence; model II adds controls for income, education and the neighbourhood immigrant share measured in adulthood. Both models include local authority fixed effects. Model I N = 2949; Model II N = 2810.

In supplementary materials section 4.2, we conduct several further tests of this result. First, in section 4.2.1, we conduct a sensitivity analysis (Cinelli and Hazlett 2002) and show

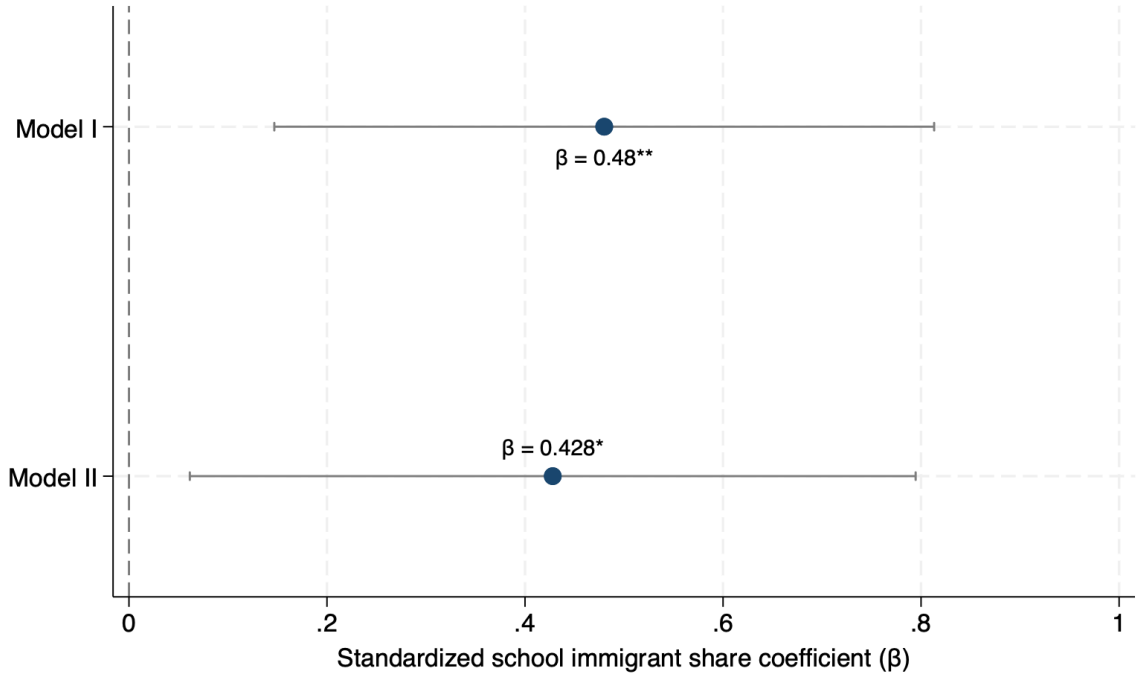
that our results are robust to unobserved confounding: an unobserved confounder 3x stronger than parental education (the adolescent control with the largest point estimate) would not explain away the association between adolescent context and adulthood immigration attitudes. Second, section 4.2.2 demonstrates that this result is robust to inclusion of several additional controls (marriage, having children, residential mobility, and occupational class). Third, some evidence suggests that the dynamics of immigration attitudes in large urban centres are distinct from other types of residential context (Maxwell, 2019). In section 4.2.3 we show that this relationship is robust to excluding sample members living in London LSOAs. Finally, in section 4.2.4 we show that our results are robust to dropping the school-length sample restriction imposed on the main models.

Finally, in section 4.2.5, we provide further evidence against the possibility that this result is driven by parental selection effects with a placebo test. If parental SES was influencing residential selection into immigrant-dense neighbourhoods beyond the observable characteristics included in our models, we would expect that our neighbourhood immigrant share measure would predict alternative, SES-driven outcomes. One such outcome is election turnout, which prior research shows is influenced by parental SES (Gidengil et al. 2016). At wave 9, respondents provide (self-reported) information on whether they voted in the most recent UK general election (held in 2019). We use this variable as a placebo outcome, regressing it on LSOA immigrant shares and the same set of controls used in model I and II in our main model specification. Results provide no evidence that neighbourhood immigrant shares are associated with the outcome, increasing confidence that the observed relationship between immigrant shares and later-life immigration attitudes are free from unobserved, parental SES-driven confounding.

School Analysis

Next, we turn to the school-level analysis. Results are presented below in figure 2, with complete model results presented in supplementary materials table 5. The school-level language variable is beta-standardized prior to inclusion in the model, meaning that the coefficient represents the change in support for immigration associated with an increase of 22.87pp in the share of the school population for which English is not their first language. As with the neighbourhood-level models, both models provide evidence that contact with immigrants during adolescence has long-term consequences for immigration in adulthood. Individuals who attended schools with a higher share of pupils for whom English was not their first language are more supportive of immigration 18 years later (relative to individuals who attended less diverse schools), although the size of this association is smaller than in the LSOA models.

Figure 2: The regression of support for immigration on adolescent school-level immigrant share



Notes: The plot reports coefficient estimates and 95% CIs. Model I controls for gender, school characteristics and parental controls in adolescence; model II adds controls for income, education and the neighbourhood immigrant share measured in adulthood. Both

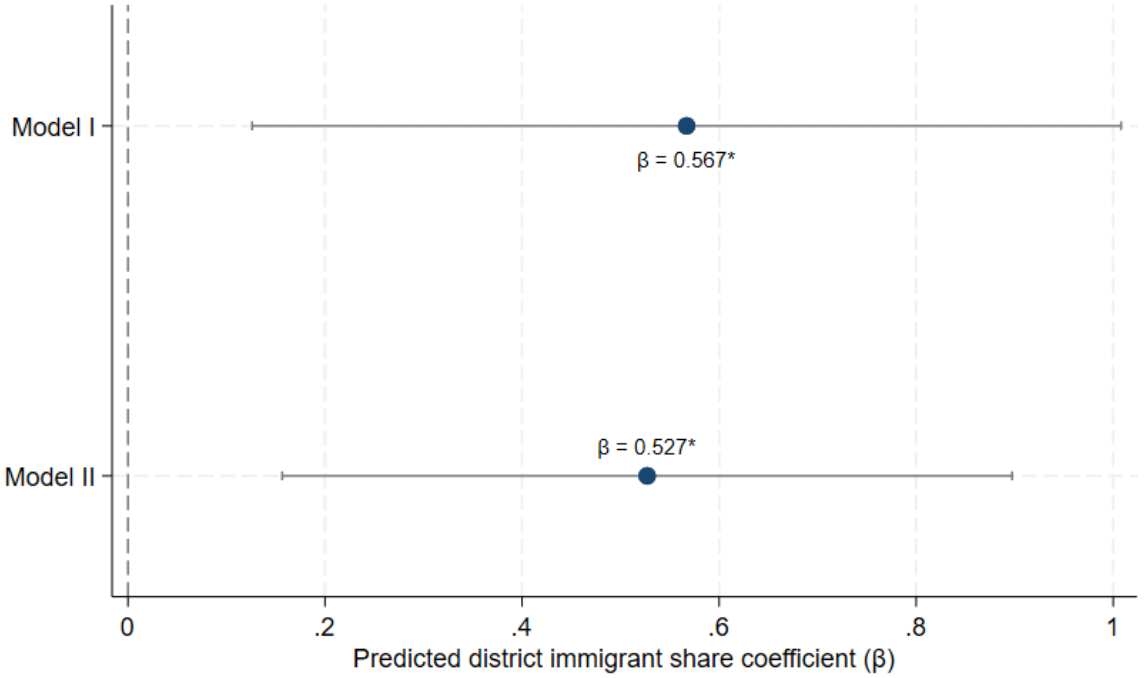
models include local authority fixed effects. Model I N = 2785; Model II N = 2648.

Shift-Share Analysis

Finally, we turn to our shift-share analysis. We display the relationship between our instrument and actual district-level shares in table 6.1 and figure 6.1 of the supplementary materials. This relationship is strong (F-statistic = 697.7) and robust to the inclusion of additional district-level controls. Next, we regress attitudes to immigration on district-level immigrant shares, instrumented by our shift-share variable, and present the second stage results below in figure 3. The immigrant share variable is again standardized, meaning that the coefficient corresponds to an increase of 1 *SD* (9.32pp) in the district-level immigrant share. These 2SLS estimates, presented in figure 3, indicate once again that living in an area with a higher share of immigrants during adolescence is associated with greater support for immigration in adulthood, and strengthen our claim that this represents a causal effect of intergroup contact rather than a byproduct of residential sorting.

The magnitude of this estimate is meaningful. Based on predicted outcome values derived from Model II, an individual who lives in an age-14 district at the 90th percentile for immigrant share (14.18%) is predicted to be 0.66 points (on the 0-10 outcome scale) more supportive of immigration in adulthood than an individual who lives in an age-14 district at the 10th percentile for immigrant share (5.3%). An individual who resides in an extremely immigrant-dense district at age 14 (the 99th percentile, or an immigrant share of 33.38%) is predicted to be 1.75 points more supportive of immigration in adulthood than one in a 10th percentile district at age 14.

Figure 3: The second-stage regression of support for immigration on adolescent neighbourhood immigrant share



Notes: The plot reports coefficient estimates and 95% CIs. Model I controls for gender and parental controls in adolescence; model II adds controls for income, education and the neighbourhood immigrant share measured in adulthood. Both models include local authority fixed effects. Model I N = 3316 ; Model II N = 3167.

Our instrument meets the usual criteria for instrument strength, with both Cragg-Donald F (1441.26 in Model I; 1398.98 in model II) and the cluster-robust Kleibergen-Paap F (16.91 in model I; 18.32 in model II) exceeding the Stock and Yogo (2005) 10% critical value for a single instrument and endogenous regressor in both models. Given this, our estimated effects are unlikely to be upwardly biased as a result of weak instruments (see Lal et al. 2024). Furthermore, in supplementary materials section 6.3.1, we show that results are robust to controlling for additional district-level characteristics measured during the 1961 Census, at the same point as our historic shares, suggesting that this result is not driven by persistent confounding.

As estimates derived from shift-share designs can be sensitive to model specification choices, we conduct some additional robustness tests. First, we show that our results are robust to excluding districts with the highest immigrant shares and to excluding all London districts (where many immigrants are concentrated), suggesting our results are not driven by outliers (section 6.3.2). Results are also consistent when we limit our sample to only those districts with the best fit between synthetic and actual district boundaries (section 6.3.3).

Discussion and Conclusion

This article presented evidence that, in Britain, White individuals who grow up in proximity to immigrants are more supportive of immigration up to two decades later. Evidence for this association was found at two levels of analysis, both in the neighborhood and at school, and remained robust after parental SES, contemporaneous neighborhood characteristics, and individual controls were taken into account. The magnitude of the association is meaningful: a 10% increase in the neighbourhood (LSOA) immigrant share at age 14 is predicted to correspond with a 0.4-point increase in support for immigration (measured on a 0-10 scale) in adulthood. The lasting effect of local immigrant shares on attitudes was further confirmed in an instrumental variable model addressing residential self-selection among immigrants.

These results put contact theory and the impressionable years hypothesis to the test, providing strong support for both. Our findings are in line with a stream of research that argues that political attitudes develop at a relatively young age only to exhibit limited change later in life (e.g. Devine & Valgarðsson, 2024; Dinas, 2010; Jennings & Niemi, 1978; Kustov et al., 2021). We showed that contact with out-groups in the context of the neighborhood and school environments during adolescence plays a role in developing more tolerant attitudes to those groups. This evidence adds to a wide body of empirical work supporting contact

theory's premise that proximity to outgroups fosters tolerance (Binder et al., 2009; van Heerden and Ruedin, 2019; Khali & Naumann, 2022; Vezzali & Stathi, 2021; Vasilopoulos et al. 2022). But while most studies document this association by focusing on the role of neighborhood characteristics in adulthood, we show that the environments in which people spent their impressionable years play an independent role in attitude formation. Further, taking the impressionable years as an analytical perspective offers new insights into the conditions under which proximity to out-groups might translate into increased tolerance as opposed to increased prejudice, as posited by ethnic threat. Indeed, while rapid inflows of immigrants in a local area may trigger negative attitudes among the majority (Enos, 2017; Hangartner et al., 2019), the social relationships involved in such dynamics differ from the strong and durable attachments likely to form during the period of adolescence. Future research could explore whether growing up in ethnically homogenous local communities that experienced a sudden influx of foreigners - conditions under which Allport's assumptions for positive contact are violated - results in less favorable attitudes towards immigrants.

These findings come with important societal and political implications regarding the long-term consequences of increasing ethnoracial diversity in advanced democracies. As the demographic composition of these countries shifts due to past and current international migration, residential environments are also transformed, with an increasing number of formerly low diversity neighbourhoods now experiencing ethnic mixing (e.g. Catney et al., 2022). Our results indicate that growing up in these ethnically diverse neighborhoods could foster increased tolerance toward ethnic outgroups that may persist for decades. This is likely to further fuel political conflict around immigration between younger cohorts who experienced diversity and older generations who did not, adding to the existing generation gap on attitudes toward immigration (McLaren et al., 2021). The findings also point to the implications of spatial inequality for political attitudes, joining the broad literature on

neighborhood effects, and suggesting that reductions in residential segregation between immigrants and natives could provide greater opportunities for contact. Recent research from the UK suggests that ethnoracial residential segregation has in fact been declining over several decades (Catney 2020), with potential positive long-term consequences for inter-group attitudes.

While we find robust effects of local immigrant shares in neighborhoods and schools consistent with contact theory, there are nonetheless limitations inherent to our data and research design which require us to be cautious with our causal interpretation. Specifically, whilst we have taken several steps to limit concerns with confounding driven by parental residential selection based on immigration preferences - including parental SES controls, higher-order geographic fixed effects, a placebo test and school-level models - we cannot completely rule out this concern. That said, it is important to note that the sensitivity analysis (supplementary materials section 4.2.1) indicates that an unobserved confounder would have to be very large, relative to benchmarked covariates, to explain away our estimates for the neighbourhood immigrant share. Further, prior research from the U.K. suggests that prior parental attitudes towards immigration are unlikely to be driving our findings: Kaufmann and Harris (2015) and Kaufmann (2023) find no evidence that anti-immigrant sentiment shapes White's residential mobility in or out of immigrant areas. Nonetheless, future research which exploits quasi-random assignment of residential context during the impressionable years is necessary to entirely rule out this concern.

A final limitation of our findings is that they come from one generation in one country. People born in 1990 grew up in a time of relative economic prosperity in a multicultural Britain, where levels of ethnic diversity were rapidly growing[2] (Rees and Butt

[2] In 2001, the last census before the period of our data collection, the share of the population of England identifying as an ethnic group other than White was 9%, up from 6% in 1991 (Rees and Butt 2004).

2004) and acceptance for such diversity was higher compared to older generations (McLaren et al., 2021). The historical context of our data collection also precedes the 2004 EU enlargement which resulted in a stark increase of international migration flows from Eastern Europe, shown to have triggered support for the far-right in certain areas of the UK (Drinkwater et al. 2009; Pupaza and Wehner 2023). Future research could thus extend this analysis into the past, by looking at the effects of contact on older generations who were raised in a more ethnically homogenous national context, and into the future, by exploring how the attitudes of more recent generations responded to the increasing diversification of the demographic landscape of the UK in the first quarter of the twenty-first century.

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Supplementary Materials for The Ties that Bind: Growing Up with Immigrants and Later Life Attitudes toward Immigration

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1 Attrition Analysis

As our analysis is built around a cohort study, non-random attrition is a potential concern. At wave 9, 7,279 members of the original cohort of 15,770 completed the survey. In this section, we attempt to understand whether this attrition might bias our results. Specifically, we compare respondent information measured at age 14 between those individuals who remained in the sample in adulthood (i.e. at wave 9, when our dependent variable in the above models is measured) and those individuals who drop out of the sample before wave 9.

Specifically, between the two groups, we conduct two-tailed t-tests to compare the: neighbourhood (LSOA) immigrant share at age 14; the percentage who are female; the mean parental household income; the percentage for whom their main parent has a degree; and the percentage for whom their main parent works in a managerial & professional occupation. These comparisons are presented below in Table 1. The standard errors and the N for each group within each comparison is presented in brackets; the p-values for the two-tailed t-tests are presented in the third column of Table 1.

Results show that, whilst the two groups differ on socio-demographics, they do not differ in terms of the immigrant share of their residential environments. We control for these sociodemographic correlates of missingness to minimise potential bias (Arel-Bundock and Pelc, 2018). Combined with the use of non-response weights, we are confident that this strategy is sufficient to address concerns with non-response bias.

Table 1: Attrition analysis

	Present at W9	**Absent at W9**	*P-value*
NH immigrant share	6.36 (SE = 0.093; $n = 5254$)	6.17 (SE = 0.086; $n = 5254$)	0.122
Gender	0.561 (SE = 0.007, $n = 4987$)	0.407 (SE = 0.007, $n = 5089$)	< 0.000
Parental HH income	3.18 (SE = 0.006, $n = 4086$)	3.08 (SE = 0.007, $n = 4048$)	< 0.000
Main parent has degree	0.136 (SE = 0.005, $n = 4998$)	0.079 (SE = 0.004, $n = 5185$)	< 0.000
Main parent is homeowner	0.797 (SE = 0.006, $n = 4998$)	0.673 (SE = 0.007, $n = 5200$)	< 0.000
Main parent social class (% managerial and professional)	0.362 (SE = 0.007, $n = 4895$)	0.267 (SE = 0.006, $n = 5080$)	< 0.000

2 Immigration Attitudes in Britain

The dependent variable in our analysis measures beliefs about the economic consequences of immigration (specifically, the question asks: "would you say it is generally bad or good for Britain's economy that migrants come to Britain from other countries?"). We assume that this measure taps a broader attitude toward or support for immigration in general. Whilst we cannot validate this assumption in our data due to the absence of alternative measures of attitudes toward immigration, we can do so with other British data measured at the same time.

Specifically, in the British Election Study Internet Panel, there are measures of economic and cultural attitudes to immigration, along with general support for increasing or limiting immigration to Britain. Two BESIP waves - waves 23 and 24 - are fielded in 2022, the same year that our outcome variable is measured in the Next Steps survey. We test the correlations between these measures to provide support for our assumption that our outcome variable, which measures economic attitudes ahout immigration, is likely to capture support for immigration in general.

At the first of these BESIP waves, fielded in May 2022, we have both measures of economic attitudes to immigration ("Do you think immigration is good or bad for Britain's economy?") and a general measure of support for immigration ("Some people think that the UK should allow many more immigrants to come to the UK to live and others think that the UK should allow many fewer immigrants. Where would you place yourself on this scale?"). The correlation between these two measures is 0.82, suggesting they tap a shared dimension of immigration attitudes.

The second of these waves, fielded in December 2022, includes both the same economic measure and a measure of cultural attitudes to immigration ("Do you think that immigration undermines or enriches Britain's cultural life?"). The correlation between these two measures is 0.83, providing further support for a single dimension of immigration attitudes.

Finally, we test formally whether all three measures load onto a single dimension of immigration attitudes. To do so, we use data from wave 27 (June 2024) which is the next wave at which all three immigration questions are asked. Correlations between the three items range from 0.79 to 0.83 and they form a scale with high reliability ($\alpha = 0.9$). Furthermore, confirmatory factor analysis indicates that all three items load onto a single latent dimension, with standardized loadings ranging from $\lambda = 0.87$ to 0.92. Given this, we are confident that the outcome measure used in our analysis is a valid proxy for broader attitudes toward immigration.

3 List of Model Variables

In this section, we detail the measurement of right-hand side variables included in our regression models.

3.1 Neighbourhood Models

The key explanatory variable in these models is the share of the neighbourhood-level population that are immigrants. We measure this with data at the LSOA level, a census geography which contains between 1000-3000 residents. We take LSOA-level immigrant shares from the 2001 Census, which was conducted three years prior to Next Steps wave 1.

For our neighbourhood-level analysis, we fit two models which each include a different set of control variables. Model I includes controls at the LSOA-level and sociodemographics measured in adolescence. Specifically, we include measures of the unemployment rate and population density within the respondent's own LSOA at age 14. We derive these measures from the 2001 Census (when our immigrant share variable is also measured). We also control for gender, parental education (a dummy for whether the respondent's 'main parent' possesses a degree)[1], parental homeownership (a dummy for whether the respondent's parents own their home), parental household income (log); and parental social class, using the NS-SeC 3 classification for the respondent's 'main parent' (Managerial and Professional, Intermediate, and Routine/Semi-Routine, along with an additional fourth category for those who have never worked or are long-term unemployed).

Neighbourhood model II includes additional controls measured in adulthood (i.e at wave 9/age 32, measured concurrently with the outcome variable). Specifically, we control for respondent education (a dummy for possessing a degree) and household income (measured in quartiles). We also control for the neighbourhood (LSOA) immigrant share where the respondent lives in adulthood using data from the 2021 Census, measured 1 year prior to Wave 9.

3.2 School Models

The key explanatory variable in our school-level models is the share of the pupil population for whom English is not their first language. We do not have school-level country of birth data in the Next Steps survey, so we use language data as a proxy.

As with the neighbourhood-level models, we fit two models. In model I we include school-level controls and sociodemographics measured in adolescence. To measure school-level deprivation, we control for the share of school pupils who are eligible for free school meals, a means-tested benefit which is provided to children from low-income households. To measure school performance, we control for the KS3 value added score of each school,

[1] At wave 1, Next Steps divided parents in two-parent households into a main parent - who was 'most involved in the young person's education' - and a second parent. Due to the definition of the 'second parent' at wave 1, large numbers of second parents in two-parent households were not interviewed at wave 1, resulting in large amounts of missing data on second parent variables, including the sociodemographic measures we use here. Consequently, we have taken the decision to focus on main parent information for education and social class.

which measures the average distance between the actual performance of school pupils at age 14 and their expected performance based on test results at age 11. We also control for gender, parental education, parental homeownership, parental household income and parental social class. The measurement of these sociodemographic variables is described above in section 3.1.

In school model II, we add controls measured in adulthood (in wave 9, when the cohort members are 32 years old). Specifically, we control for respondent education and household income, along with the neighbourhood (LSOA) immigrant share taken from the 2021 Census.

4 LSOA Results

4.1 Main Models

The results presented below in Table 4.1 correspond to figure 1 in the main text.

Table 4.1: The Regression of Support for Immigration on Adolescent Neighbourhood-Level Immigrant Share

VARIABLES	**Model I**	**Model II**
NH Immigrant Share (std)	0.673***	0.508***
	(0.165)	(0.144)
NH Unemployment	-0.099*	-0.050
	(0.039)	(0.035)
NH Population Density	-0.005+	-0.003
	(0.003)	(0.003)
Parent has degree	0.963***	0.542***
	(0.157)	(0.152)
Parent owns home	0.341*	0.051
	(0.148)	(0.142)
Female	-0.266*	-0.417**
	(0.104)	(0.103)
Parental income (log)	0.246	0.077
	(0.169)	(0.165)
Parent class: *Intermediate*	-0.102	-0.119
	(0.146)	(0.134)
Routine/Semi-Routine	-0.673***	-0.481***
	(0.149)	(0.138)
Never Worked/Long-Term Unemployed	-0.633	-0.477
	(0.458)	(0.460)
Degree		1.414***
		(0.110)
HH Income: 2nd Quartile		0.312
		(0.190)
HH Income: 3rd Quartile		0.643**
		(0.212)
HH Income: 4th Quartile		1.037***
		(0.194)
NH Immigrant Share (Adulthood)		0.013*
		(0.005)
Observations	2,949	2,810
R-squared	0.220	0.322

Notes: Clustered SEs (by local authority and primary sampling unit) are reported in parentheses. Coefficients for local authority dummies are not reported. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{+}p < 0.10$.

4.2 Further Analysis

4.2.1 Sensitivity Analysis

An important consideration when assessing these results is their sensitivity to unobserved confounding. We adopt the omitted variable bias framework from Cinelli and Hazlett (2020) to assess the degree to which the LSOA results presented in Table 4.1 are sensitive to omitted variables.

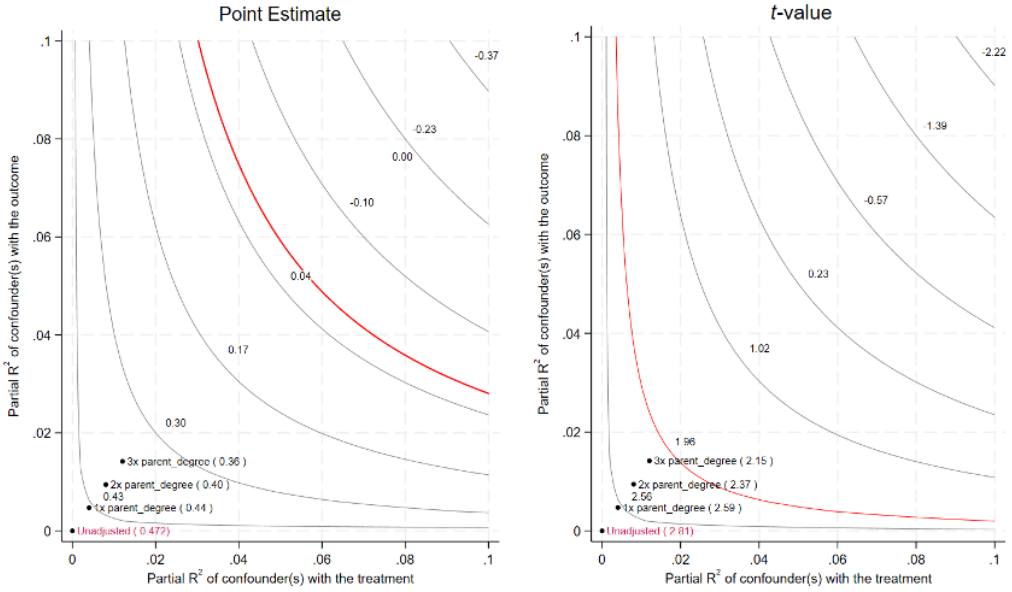
Our primary concern is parental immigration attitudes, which could both influence residential context (via residential selection) and the immigration attitudes of our cohort members (via parental socialization). Whilst we control for the sociodemographic correlates of immigration attitudes among parents, parental attitudes are unobserved, and sensitivity analysis allows us to quantify the degree to which potential unobservables threaten our findings.

We conduct our sensitivity analysis using the *sensemakr* package from Cinelli, Ferwerda and Hazlett (2024). We first calculate the robustness value for the LSOA immigrant share point estimate from model II (Table 4.1). This robustness value ($RV_{q=1}$) is 0.0543, meaning that an unobserved confounder which explains 5.43% of the residual variance of the LSOA immigrant share and the outcome (immigration attitudes) would explain away all of the observed relationship between our the two measures.

We then benchmark the effect of an unobserved confounder against that of an observed confounder. We benchmark against parental education because (a) the point estimate for parental education is the largest among observed confounders measured in adolescence (Table 4.1) and (b) parental education is, conceptually, the closest observable to the unobserved confounder which poses the greatest endogeneity concerns for our model - namely, parental attitudes to immigration.

Figure 4.2.1 below displays this sensitivity analysis visually. The left-hand panel plots the coefficient estimate for the LSOA immigrant share in the unadjusted model and in hypothetical models in which an unobserved confounder, benchmarked against parental education, is included. The right-hand panel displays the estimated t-value from the unadjusted model and from models including a benchmarked unobserved confounder. This plot shows that even a confounder 3x as strong as parental education would not alter our conclusions: we would still observe a positive association between neighbourhood immigrant shares in adolescence and immigration attitudes in adulthood, and reject the null hypothesis of no effect (at $p < 0.05$), if a confounder of this strength was included.

Figure 4.2.1: Sensitivity analysis for the regression of immigration attitudes on adolescent neighbourhood immigrant shares



Note: Model $N = 2833$.

4.2.2 Additional Controls

Next, we test the robustness of our findings to the inclusion of additional control variables, measured in adulthood. Specifically, we add dummies for employment, being a parent, marriage, homeownership, social class (a dummy for managerial or professional occupations) and residential mobility. The latter is measured with a dummy which equals 1 if the respondent resides in the same LSOA in adulthood as they did in adolescence.[2] Results are presented below in Table 4.2.2.

Table 4.2.2: The Regression of Support for Immigration on Adolescent Neighbourhood-Level Immigrant Share, Additional Controls

VARIABLES	**Additional Controls**
NH Immigrant Share (std)	0.464** (0.143)
NH Unemployment	-0.028 (0.035)
NH Population Density	-0.003 (0.002)
Parent has degree	0.519*** (0.152)
Parent owns home	-0.050 (0.145)
Female	-0.347*** (0.104)
Parental income (log)	0.012 (0.162)
Parent class: *Intermediate*	-0.082 (0.135)
Routine/Semi-Routine	-0.403** (0.139)
Never Worked/Long-Term Unemployed	-0.431 (0.450)
Degree	1.179*** (0.115)
HH Income: 2nd Quartile	0.133 (0.200)
HH Income: 3rd Quartile	0.312 (0.221)
HH Income: 4th Quartile	0.591** (0.218)
NH Immigrant Share (Adulthood)	0.009+ (0.005)
Not moved LSOA	-0.479** (0.155)
Parent	-0.478*** (0.106)
Married	0.069 (0.147)
Managerial & Professional	0.410** (0.125)
Homeowner	0.200 (0.123)
In work	0.263 (0.210)
Observations	2,801
R-squared	0.314

Notes: Clustered SEs (by local authority and primary sampling unit) are reported in parentheses. Coefficients for local authority dummies are not reported. ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$, +$p < 0.10$.

[2]Around 20% of the sample reside in the same LSOA in adulthood as in adolescence. This is likely to be a slightly conservative estimate, as the mobile group includes those respondents who have not relocated but who reside in LSOAs where the codes changed between censuses ($< 5\%$ of total LSOAs).

4.2.3 Excluding London

Recent analysis of the geography of immigration attitudes suggests that immigration attitudes in large urban centers are distinct from other types of residential context (Maxwell, 2019). London is by far the largest urban centre in England - the 2001 Census recorded the London population as 7,172,091, whereas the second-largest city by population was Birmingham with a population of 977,087. Given the size of London relative to the rest of England, one possibility is that London is driving the relationship observed in our main models. To test this possibility, we estimate models that exclude respondents living in London LSOAs. Results are presented below in Table 4.2.3 and indicate that the relationship between neighborhood immigrant share in adolescence and attitudes toward immigration in adulthood is robust to excluding London.

Table 4.2.3: The Regression of Support for Immigration on Adolescent Neighbourhood-Level Immigrant Share, Excluding London

VARIABLES	**Model I**	**Model II**
NH Immigrant Share (std)	0.640*** (0.167)	0.462** (0.150)
NH Unemployment	-0.079* (0.039)	-0.038 (0.036)
NH Population Density	-0.005+ (0.003)	-0.003 (0.003)
Parent has degree	0.877*** (0.157)	0.468** (0.155)
Parent owns home	0.272+ (0.153)	0.032 (0.147)
Female	-0.292** (0.109)	-0.452*** (0.108)
Parental income (log)	0.278 (0.180)	0.089 (0.177)
Parent class: *Intermediate*	-0.056 (0.152)	-0.072 (0.139)
Routine/Semi-Routine	-0.764*** (0.150)	-0.560*** (0.140)
Never Worked/Long-Term Unemployed	-0.724 (0.490)	-0.508 (0.498)
Degree		1.358*** (0.115)
HH Income: 2nd Quartile		0.255 (0.195)
HH Income: 3rd Quartile		0.548* (0.215)
HH Income: 4th Quartile		0.960*** (0.196)
NH Immigrant Share (Adulthood)		0.019*** (0.005)
Observations	2,767	2,638
R-squared	0.220	0.294

Notes: Clustered SEs (by local authority and primary sampling unit) are reported in parentheses. Coefficients for local authority dummies are not reported. ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$, +$p < 0.10$.

4.2.4 Excluding School-Length Restriction

The main LSOA models impose a sample restriction - we limit the model sample to respondents who have studied at the same school since 2001. We do this based on the assumption that an absence of school change is likely to imply stability in residential context, helping to validate the assumption that the respondent resided in the neighborhood for a prolonged period (and thus experienced contact with local residents). In Table 4.2.4 we re-estimate the main LSOA models, dropping this sample restriction. The results are robust to this change.

Table 4.2.4: The Regression of Support for Immigration on Adolescent Neighbourhood-Level Immigrant Share, Excluding School-Length Restriction

VARIABLES	**Model I**	**Model II**
NH Immigrant Share (std)	0.561*** (0.155)	0.393** (0.138)
NH Unemployment	-0.107** (0.039)	-0.064+ (0.035)
NH Population Density	-0.003 (0.003)	-0.001 (0.002)
Parent has degree	1.049*** (0.141)	0.560*** (0.138)
Parent owns home	0.380** (0.132)	0.068 (0.127)
Female	-0.308** (0.096)	-0.432*** (0.093)
Parental income (log)	0.168 (0.151)	-0.021 (0.148)
Parent class: *Intermediate*	-0.137 (0.132)	-0.147 (0.125)
Routine/Semi-Routine	-0.625*** (0.134)	-0.423*** (0.122)
Never Worked/Long-Term Unemployed	-0.702+ (0.390)	-0.606 (0.383)
Degree		1.472*** (0.102)
HH Income: 2nd Quartile		0.321+ (0.180)
HH Income: 3rd Quartile		0.634** (0.191)
HH Income: 4th Quartile		0.973*** (0.178)
NH Immigrant Share (Adulthood)		0.016*** (0.005)
Observations	3,450	3,294
R-squared	0.209	0.291

Notes: Clustered SEs (by local authority and primary sampling unit) are reported in parentheses. Coefficients for local authority dummies are not reported. ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$, +$p < 0.10$.

4.2.5 Placebo Test

In Table 4.2.5 we present the results from a placebo test regression of 2019 general election turnout on neighbourhood (LSOA) immigrant shares in adolescence. Results provide no evidence that neighbourhood immigrant shares impact self-reported turnout propensity later in life.

Table 4.2.5: The Regression of 2019 General Election Turnout on Adolescent Neighbourhood-Level Immigrant Share

VARIABLES	**Model I**	**Model II**
NH Immigrant Share (std)	0.038 (0.042)	0.036 (0.040)
NH Unemployment	-0.028*** (0.006)	-0.017* (0.007)
NH Population Density	-0.001 (0.000)	-0.000 (0.000)
Parent has degree	0.099*** (0.029)	0.050+ (0.030)
Parent owns home	0.142*** (0.030)	0.088** (0.030)
Female	-0.062** (0.019)	-0.078*** (0.020)
Parental income (log)	0.023 (0.028)	0.007 (0.028)
Parent class: *Intermediate*	0.026 (0.027)	0.022 (0.027)
Routine/Semi-Routine	-0.071** (0.027)	-0.041 (0.026)
Never Worked/Long-Term Unemployed	-0.048 (0.070)	-0.009 (0.071)
Degree		0.194*** (0.022)
HH Income: 2nd Quartile		0.077* (0.036)
HH Income: 3rd Quartile		0.142*** (0.037)
HH Income: 4th Quartile		0.211*** (0.039)
NH Immigrant Share (Adulthood)		-0.000 (0.001)
Observations	3,027	2,858
R-squared	0.184	0.242

Notes: Clustered SEs (by local authority and primary sampling unit) are reported in parentheses. Coefficients for local authority dummies are not reported. ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$, +$p < 0.10$.

5 School Results

The results presented below in Table 5.1 correspond to figure 2 in the main text.

Table 5.1: The Regression of Support for Immigration on Adolescent School-Level Immigrant Share

VARIABLES	**Model I**	**Model II**
School % Non-English Speaking (std)	0.480** (0.170)	0.428* (0.187)
% Free School Meals	-0.012 (0.010)	-0.002 (0.010)
KS3 Value Added	0.042 (0.080)	-0.051 (0.075)
Parent has degree	0.957*** (0.171)	0.473** (0.148)
Parent owns home	0.400** (0.147)	0.058 (0.141)
Female	-0.276* (0.109)	-0.421*** (0.107)
Parental income (log)	0.215 (0.183)	0.066 (0.173)
Parent class: *Intermediate*	-0.038 (0.150)	-0.086 (0.139)
Routine/Semi-Routine	-0.643*** (0.152)	-0.491*** (0.140)
Never Worked/Long-Term Unemployed	-0.750 (0.466)	-0.610 (0.472)
Degree		1.520*** (0.115)
HH Income: 2nd Quartile		0.288 (0.198)
HH Income: 3rd Quartile		0.640** (0.218)
HH Income: 4th Quartile		1.031*** (0.201)
NH Immigrant Share (Adulthood)		0.017** (0.006)
Observations	2,785	2,648
R-squared	0.207	0.292

Notes: Clustered SEs (by local authority and primary sampling unit) are reported in parentheses. Coefficients for local authority dummies are not reported. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $+p < 0.10$.

6 Shift-Share Analysis

6.1 Instrument Construction

Our shift-share instrument uses immigrant shares at the local authority district level, taken from the 1961 Census, to construct the base shares. These districts are larger than the LSOAs used in the prior neighbourhood models (there were 1297 local authority districts in England at the 1961 Census with a median population of 17,619 residents), but they are the most fine-grained geographic unit for which complete data is available, allowing us to measure immigrant shares in 1961 across the entirety of England.

Before constructing the instrument, we first construct a synthetic geography at the 2001 Census that mirrors the 1961 district boundaries. Whilst local authorities remained in use as geographic units throughout our study period, districts (and their boundaries) changed significantly from 1961 to 2001, meaning that a consistent geographic unit is not available for our use. However, this poses minimal problems for our analysis.

Using 2001 LSOAs as building blocks, we construct geographic units that mirror, as closely as possible, the geographic area covered by each local authority district in 1961. Specifically, we identify the 1961 district that shares the greatest geographic area with each 2001 LSOA and assign this 1961 district to each LSOA. Then, joining together those LSOAs that share the same 1961 district, we construct higher-level geographies from the 2001 LSOAs—which we call synthetic districts—that mirror (as closely as possible) the 1961 district boundaries. The fit between these synthetic districts and the 1961 districts which they mirror is very good: across 1961-2001 pairs, the Jaccard coefficient (the proportion of the total area covered by both units which is shared by both units) has a mean of 0.88 and a median of 0.92.

With a consistent geographic unit across our entire study period, we are able to construct our instrument. First, we define the shares for each immigrant origin group (o) in district (i) in 1961 as:

$$\alpha_i^o = \frac{M_{i,1961}^o}{M_{1961}^o} \tag{1}$$

This is the number of migrants from origin o in district i in 1961 as a proportion of the total number of migrants from origin o in England in 1961. We cluster immigrants into the following origin groups, using adapted versions of the categories provided in the 1961 census: Irish (including both individuals from Northern Ireland and the Republic of Ireland); Indian subcontinent (India, Pakistan, Bangladesh and Sri Lanka); Caribbean; an 'other Commonwealth' category containing Australia, Canada, Malaysia, New Zealand and South Africa; and an 'other foreign' category, a composite category containing immigrants from all other destinations.[3]

[3]Ideally we would use a more fine-grained categorization of origin groups. Due to a combination of the countries identified in the 1961 Census and the need for continuity between 1961 and 2001, this categorization is the most fine-grained approach we can use.

Next, for each origin population, we compute the national-level shifts, defined as:

$$\Delta M^{o}_{2001} = M^{o}_{2001} - M^{o}_{1961} \tag{2}$$

So, for each migrant origin group o, we compute the change in the total population in England between 1961 and 2001. Combining these together, we predict the immigrant in-flows to each synthetic district in 2001 as:

$$\Delta M_{i,2001} = \sum_{o} \left(\frac{M^{o}_{i,1961}}{M^{o}_{1961}} \right) \Delta M^{o}_{2001} \tag{3}$$

We then scale these district-level immigrant counts by the 2001 district total population. To avoid dividing by an endogenous variable, we use predicted 2001 district populations, constructed by multiplying 1961 district population shares by the national-level 1961 to 2001 population shift. 25 LADs are excluded from the instrument due to missing data on immigrant counts for at least one of the origin groups in the 1961 census.

Table 6.1 shows that the first-stage relationship between the district-level immigrant shares and the predicted shares based on our instrument is strong and remains robust to controlling for region fixed effects and a set of baseline district-level characteristics measured in the 1961 Census, when our base shares are measured (population density, the graduate rate, the share of workers in industry and the male unemployment rate). We also report the conventional F-statistic from each of these models.

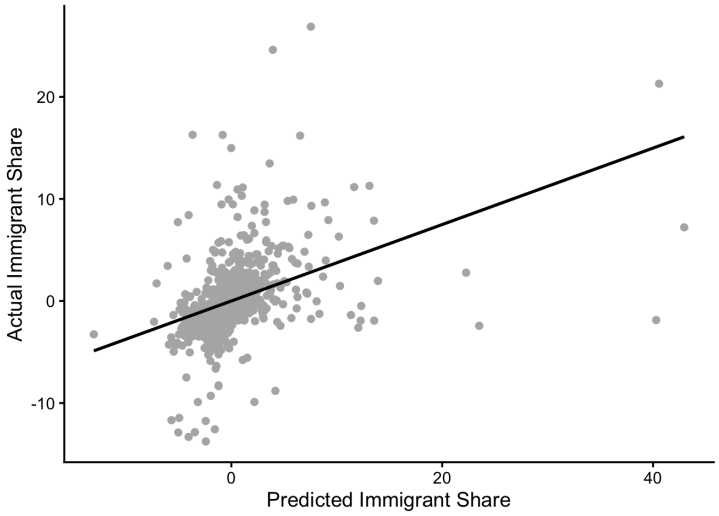
Figure 6.1 plots the first stage relationship from the model which includes both region fixed effects and baseline controls.

Table 6.1: First Stage Relationship

VARIABLES	**Model I**	**Model II**	**Model III**
Predicted Immigrant Share	0.92***	0.532***	0.375***
	(0.035)	(0.028)	(0.025)
Region FE		YES	YES
Baseline Controls			YES
Observations (Districts)	1272	1272	1135
F-statistic	697.7	318.2	331.6
R-squared	0.354	0.714	0.803

Notes: The unit of analysis for this model is 1961 local authority districts. Baseline controls, measured at the district level, are population density, the graduate rate, the share of workers in industry and the male unemployment rate. Coefficients for region dummies are not reported. ***$p < 0.001$.

Figure 6.1: First Stage Relationship



Notes: The figure plots the partial relationship between the shift-share instrument and the actual immigrant share in 2001 across local authority districts after residualizing both variables with respect to government office region fixed effects and the baseline controls. Each point represents one district. The solid line shows the fitted OLS regression line.

6.2 Main Model Results

The results presented below in Table 6.2 correspond to figure X in the main text.

Table 6.2: The Regression of Support for Immigration on Predicted Adolescent District-Level Immigrant Share

VARIABLES	**Model I**	**Model II**
Predicted District Immigrant %	0.567* (0.225)	0.527** (0.189)
Parent has degree	1.123*** (0.143)	0.571*** (0.136)
Parent owns home	0.565*** (0.135)	0.179 (0.133)
Female	-0.348*** (0.098)	-0.486*** (0.091)
Parental income (log)	0.194 (0.148)	-0.041 (0.140)
Parent class: *Intermediate*	-0.210 (0.128)	-0.210+ (0.125)
Routine/Semi-Routine	-0.850*** (0.127)	-0.495*** (0.120)
Never Worked/Long-Term Unemployed	-0.643 (0.407)	-0.433 (0.394)
Degree		1.683*** (0.097)
HH Income: 2nd Quartile		0.377* (0.186)
HH Income: 3rd Quartile		0.684*** (0.202)
HH Income: 4th Quartile		0.956*** (0.191)
Constant	5.528*** (0.533)	5.414*** (0.531)
Observations	3,316	3,167
Kleibergen-Paap F	16.908	18.316
R-squared	0.091	0.195

Notes: Clustered SEs (by district) are reported in parentheses. Coefficients for region dummies are not reported. ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$, +$p < 0.10$.

6.3 Further Analysis

6.3.1 District-Level Controls

Causal identification rests on the assumption of the exogeneity of the shares. To strengthen this assumption, we estimate models that additionally control for district-level characteristics that correlate with the 1961 immigrant shares (also measured at the 1961 Census), on top of the controls included in the main model specification.

We begin by testing the association between 1961 immigrant shares and other district-level factors that potentially correlate with the shares. Using districts as the unit of analysis and restricting the model to those districts where Next Steps respondents reside, we regress 1961 immigrant shares on four district-level variables: population density, the share of the working age population employed in production/manufacturing, the male unemployment rate (focusing only on men due to the relative rarity of women working at this time point) and the graduate rate (the share of workers who finished education at aged 20 or over). These measures are all derived from 1961 Census small area estimates (CITE), meaning they are measured at the same point as the immigrant shares used to construct the instrument.

Results from this model are presented below in Table 6.3.1a. We see that population density and the graduate rate are positively associated with the district-level immigrant share.

Table 6.3.1a: The Regression of District-Level Historic Immigrant Shares on Additional District-Level Characteristics

	VARIABLES
Population Density (std)	1.951*** (0.104)
Production Share	-0.013 (0.008)
Male Unemployment Rate	0.241 (0.157)
Graduate Rate	1.825*** (0.411)
Observations (Districts)	645
R-squared	0.492

Notes: The unit of analysis for this model is 1961 local authority districts. Coefficients for region dummies are not reported. The coefficients for the production share, male unemployment rate and graduate rate correspond to percentage point increases in the corresponding predictor, whilst population density is measured in people per KM^2 and is beta-standardized (a 1-unit increase corresponds to an increase of 2097 people per KM^2). $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{+}p < 0.10$.

The observed association between historic immigrant shares and population density and the graduate rate point to potential violations of the exogeneity assumption if these characteristics exert persistent influence on local conditions. Given this, we estimate an additional version of our shift-share analysis which controls for these two further district-level characteristics. Results are presented below in Table 6.3.1b. Models I and II also contain the same set of controls included in our main shift-share models (in Table 6.2), but we do not report their coefficient estimates here.

Results from these models indicate that our conclusions do not change when we include these historic controls. Whilst the estimate for the immigrant share in model 1 is now only statistically significant at $p < 0.1$, the consistency of the point estimates across model specifications mean that we interpret these results as further evidence in support of a relationship between immigrant shares in adolescence and attitudes in adulthood.

Table 6.3.1b: The Regression of Support for Immigration on Predicted Adolescent District-Level Immigrant Share, with 1961 District Controls

VARIABLES	**Model I**	**Model II**
Predicted District Immigrant %	0.534+ (0.32)	0.602* (0.276)
Population Density	-0.000 (0.000)	-0.000 (0.000)
Graduate Rate	0.11 (0.115)	-0.011 (0.109)
Adolescent Controls	YES	YES
Adulthood Controls	NO	YES
Region FE	YES	YES
Observations	2,760	2,646
Kleibergen-Paap F	16.531	18.647
R-squared	0.085	0.19

Notes: Clustered SEs (by district) are reported in parentheses. Coefficients for region dummies are not reported. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $+p < 0.10$.

6.3.2 Outliers

One potential concern is that the observed effect of district-level immigrant shares is driven by a few outlier districts. To address this concern, we conduct several tests. First, in table 7.1 we re-estimate model II from table 5, excluding the three districts that have the largest predicted migrant shares in 2001 (these are also the three districts that experienced the largest increase in their immigrant share from 1961 to 2001). We first exclude these one-at-a-time before excluding all three. The district immigrant share continues to have a positive effect on support for immigration even when we exclude these outlier LADs.

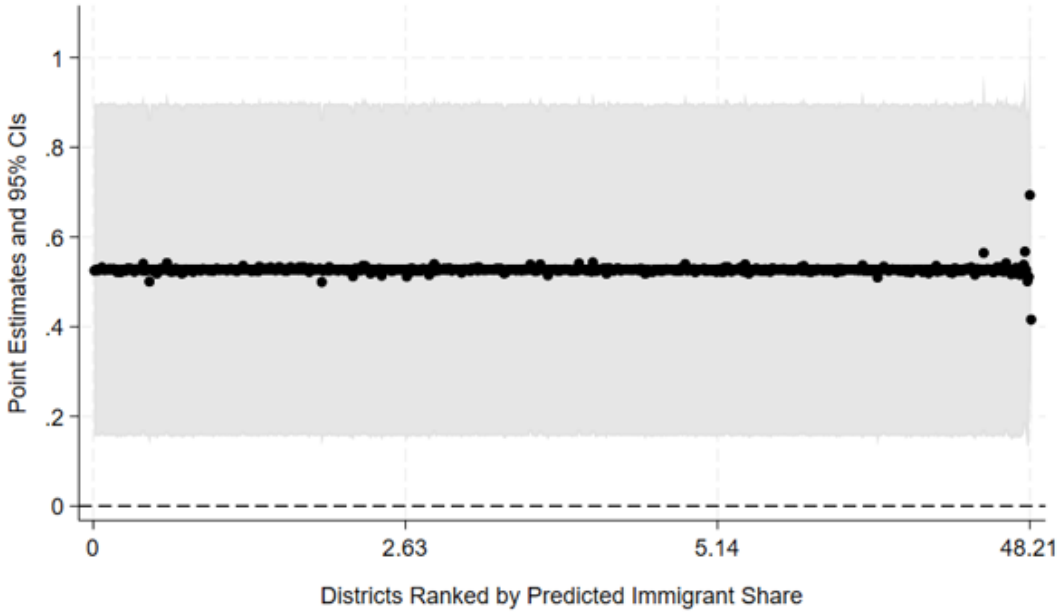
Table 6.3.2a: The Regression of Support for Immigration on Predicted Adolescent District-Level Immigrant Share, Excluding Outlier Districts

VARIABLES	**All LADs**	**Drop 1st**	**Drop 2nd**	**Drop 3rd**	**Drop All 3**
Pred. District	0.527**	0.416**	0.694***	0.511**	0.526***
Immigrant %	(0.189)	(0.140)	(0.199)	(0.193)	(0.158)
Adolescent Controls	YES	YES	YES	YES	YES
Adulthood Controls	YES	YES	YES	YES	YES
Region FE	YES	YES	YES	YES	YES
Kleibergen-Paap	18.316	23.995	7.774	16.822	91.259
Observations	3,167	3,158	3,154	3,165	3,143

Notes: Clustered SEs (by district) are reported in parentheses. Coefficients for region dummies are not reported. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $+p < 0.10$.

Next we test whether our results are sensitive to excluding individual districts from the sample. To do so we re-estimate model II in Table 6.2, dropping one district at a time. We then extract the coefficient for the predicted immigrant share from each of these models and plot each of these coefficients below in Figure 6.3.2, along with 95% confidence intervals. We see that the relationship of interest remains consistent across this test - the lower confidence interval exceeds zero on all replications.

Figure 6.3.2: Testing the Influence of Outliers



Notes: The figure plots point estimates and 95% confidence intervals for the effect of predicted district immigrant shares on support for immigration, estimating Model II in table 6.2 whilst dropping one district at a time. Districts are ordered by their predicted immigrant share (in 2001).

Finally, we test whether results are sensitive to excluding districts in London. To do so we re-estimate versions of our main shift-share models which exclude London districts. Results are presented below in Table 6.3.2b and indicate that the relationship between predicted immigrant shares and immigration attitudes remains consistent when we exclude these districts from the analysis.

Table 6.3.2b: The Regression of Support for Immigration on Predicted Adolescent District-Level Immigrant Share, Excluding London

VARIABLES	**Model I**	**Model II**
Predicted District Immigrant %	0.685* (0.305)	0.620* (0.256)
Adolescent Controls	YES	YES
Adulthood Controls	NO	YES
Region FE	YES	YES
Observations	3,123	2,984
Kleibergen-Paap F	14.104	15.762
R-squared	0.088	0.188

Notes: Clustered SEs (by district) are reported in parentheses. Coefficients for region dummies are not reported. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $+p < 0.10$.

6.3.3 Jaccard Restrictions

For our shift-share analysis, we construct synthetic local authority districts for our immigrant share measure in adolescence by aggregating LSOAs measured at the 2001 Census (we describe this procedure in detail in section 6.1). Whilst the average fit between these synthetic local authority districts and the actual local authority districts (taken from the 1961 Census) is strong (Jaccard coefficient mean = 0.88; median = 0.92), one potential concern is that the observed effect of district immigrant shares on attitudes to immigration could be upwardly biased by those districts with poor fit between their actual (1961) and synthetic (2001) boundaries. We address this concern by estimating versions of model II from our main shift-share analysis (in Table 6.2) which are limited to respondents in districts with increasingly high cut-offs for fit between synthetic and actual local authority boundaries. We conduct three versions of this test, limiting the sample to respondents in districts with Jaccard coefficients greater than 0.85, 0.9 and then 0.95 (a score of 1 indicates perfect fit). Results are presented below in Table 6.3.3. The district immigrant share estimate remains consistent across all of these models.

Table 6.3.3: The Regression of Support for Immigration on Predicted Adolescent District-Level Immigrant Share, Jaccard Restrictions

VARIABLES	**All LADs**	**Jaccard > 0.85**	**Jaccard > 0.9**	**Jaccard > 0.95**
Pred. District	0.527**	0.549**	0.421**	0.354*
Immigrant %	(0.189)	(0.194)	(0.156)	(0.160)
Adolescent Controls	YES	YES	YES	YES
Adulthood Controls	YES	YES	YES	YES
Region FE	YES	YES	YES	YES
Kleibergen-Paap	18.316	17.624	22.757	22.582
Observations	3,167	2,760	2,261	1,307
R-squared	0.195	0.203	0.200	0.213

Notes: Clustered SEs (by district) are reported in parentheses. Coefficients for region dummies are not reported. ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$, +$p < 0.10$.

References

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